Nombre del Proveedor: DELCROSA S.A.
Número de Contrato: 104688-PO-0200
Descripción del Contrato: Transformadores de Distribución
Nombre del Propietario: Compañía de Minas Buenaventura S.A.A.
Ausenco Doc. No: 104688-RP-P0200-40467- 141

# Proyecto San Gabriel – Fase 1
# VDDR: 3.03
# Equipment Data
# Tag: 2850-TX-015

DOCUMENTO CERTIFICADO
CONSTRUCCIONES ELECTROMECÁNICAS DELCROSA S.A.
ING. ENRIQUE LUJAN MENDOZA
INGENIERO ELECTRICISTA-CIP 57696
FECHA: 13-03-2023

# PERFORMANCE DATA/CURVES:
# CURVA DE EFICIENCIA

DELCROSA S.A. VºBº INGENIERÍA JORGE L. MENDEZ GIRON

| Rev | Fecha | Descripción | Elaborado | Revisado | Aprobado |
|---|---|---|---|---|---|
| A | 06/01/2023 | Emitido para revisión interna | ISO | ISO | ISO |
| B | 17/02/2023 | Emitido para revisión del cliente | GOP/ISO | GOP/ISO | ISO |
| 0 | 03/03/2023 | Emitido para fabricación | DELCROSA | AUSENCO | AUSENCO |

Ausenco

# VENDOR DATA REVIEW

**Doc. No.** 104688-RP-P0200-40467-141_Sub2.pdf

| Discipline | By | Date |
|---|---|---|
| Civil | | |
| Structural | | |
| Mechanical | | |
| Piping | | |
| Electrical | | |
| Instrumentation | | |
| Process | | |
| Drafting | | |
| Quality | | |
| Contracts/Procurement | | |

- ☐ **1 Accepted to Proceed.**
- ☐ **2 Accepted to Proceed. SUBMIT FINAL CERTIFIED COPY.**
- ☐ **3 Accepted to Proceed, except as noted. Revise & Resubmit.**
- ☐ **4 Not Accepted to Proceed. Revise & Resubmit.**
- ☐ **5 Information ONLY.**
- ☐ **6 CANCELLED.**

Review of this document has been completed to determine if the intent of the Contract Documents have generally been completed with. Such review does not relieve the Contractor of contractual obligations.

## INTRODUCCIÓN

El presente documento tiene por finalidad dar a conocer al cliente AUSENCO y a la COMPAÑÍA DE MINAS BUENAVENTURA S.A.A., la Curva de Eficiencia del Transformador a suministrar para el Proyecto “San Gabriel – 104688-PO-0200”.

La Curva de Eficiencia corresponde al siguiente Tag asignado por el cliente y el Orden de Trabajo (OT) asignado por DELCROSA S.A.:

**TABLA DE TAG y OT**

| TAG AUSENCO | OT DELCROSA |
|---|---|
| **2850-TX-015** | **149475** |

Se muestra la Curva de Eficiencia del siguiente Transformador:

| TAG | OT | POTENCIA (kVA) | AT | BT | GRUPO | DESCRIPCIÓN |
|---|---|---|---|---|---|---|
| **2850-TX-015** | **149475** | 1250 | 22900 | 480 | Dyn5 | LIQUID-IMMERSED TRANSFORMER (22.9 / 0.48 kV) – 1,250 kVA |




*Nota:*
*Se emite el documento certificado del plano APROBADO por AUSENCO según el Transmittal N° “104688-XTR-3053” remitido mediante correo el 03/03/2023.*

| | | |
|---|---|---|
| **POTENCIA:** | 1250 | kVA |
| **Pfe:** | 2600 | W |
| **Pcu:** | 14300 | W |

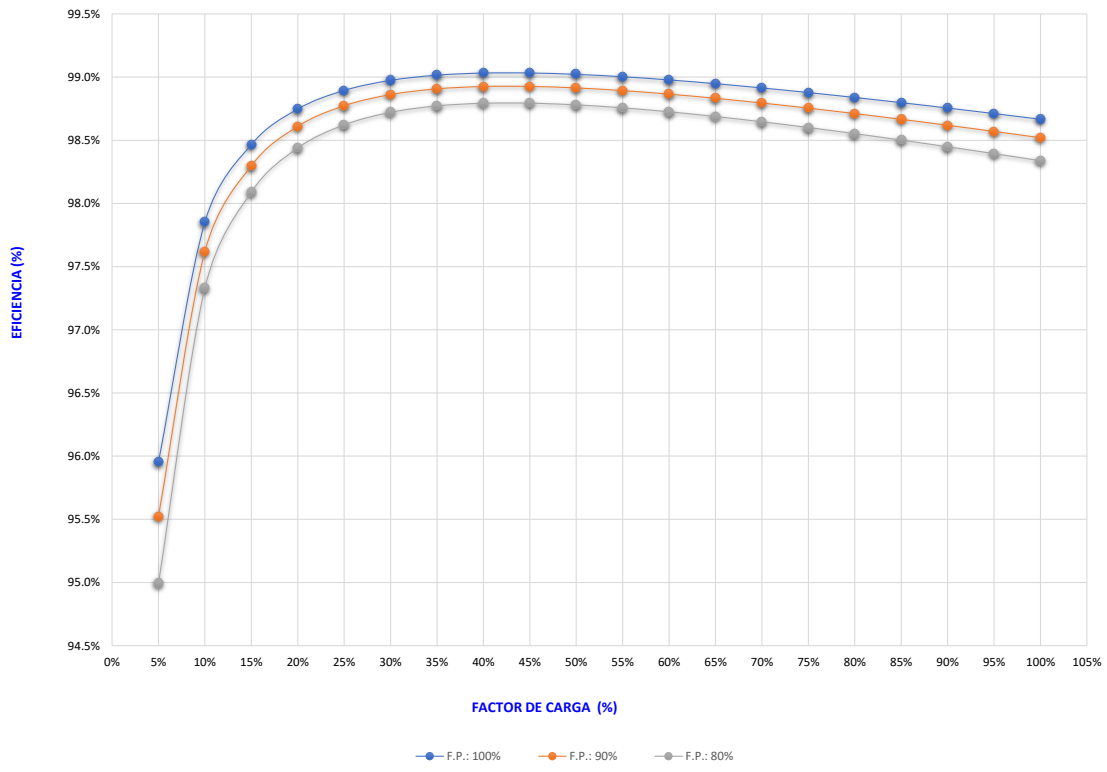
PERFORMANCE DATA/CURVES: CURVA DE EFICIENCIA
TAG: 2850-TX-015

| FACTOR DE CARGA | FACTOR DE POTENCIA | EFICIENCIA |
|---|---|---|
| 100% | 100% | 98.67% |
| 95% | 100% | 98.71% |
| 90% | 100% | 98.75% |
| 85% | 100% | 98.80% |
| 80% | 100% | 98.84% |
| 75% | 100% | 98.88% |
| 70% | 100% | 98.91% |
| 65% | 100% | 98.95% |
| 60% | 100% | 98.98% |
| 55% | 100% | 99.00% |
| 50% | 100% | 99.02% |
| 45% | 100% | 99.03% |
| 40% | 100% | 99.03% |
| 35% | 100% | 99.02% |
| 30% | 100% | 98.97% |
| 25% | 100% | 98.89% |
| 20% | 100% | 98.75% |
| 15% | 100% | 98.47% |
| 10% | 100% | 97.85% |
| 5% | 100% | 95.95% |

| FACTOR DE CARGA | FACTOR DE POTENCIA | EFICIENCIA |
|---|---|---|
| 100% | 90% | 98.52% |
| 95% | 90% | 98.57% |
| 90% | 90% | 98.62% |
| 85% | 90% | 98.67% |
| 80% | 90% | 98.71% |
| 75% | 90% | 98.75% |
| 70% | 90% | 98.79% |
| 65% | 90% | 98.83% |
| 60% | 90% | 98.87% |
| 55% | 90% | 98.89% |
| 50% | 90% | 98.91% |
| 45% | 90% | 98.93% |
| 40% | 90% | 98.93% |
| 35% | 90% | 98.91% |
| 30% | 90% | 98.86% |
| 25% | 90% | 98.77% |
| 20% | 90% | 98.61% |
| 15% | 90% | 98.30% |
| 10% | 90% | 97.62% |
| 5% | 90% | 95.52% |

| FACTOR DE CARGA | FACTOR DE POTENCIA | EFICIENCIA |
|---|---|---|
| 100% | 80% | 98.34% |
| 95% | 80% | 98.39% |
| 90% | 80% | 98.45% |
| 85% | 80% | 98.50% |
| 80% | 80% | 98.55% |
| 75% | 80% | 98.60% |
| 70% | 80% | 98.65% |
| 65% | 80% | 98.69% |
| 60% | 80% | 98.73% |
| 55% | 80% | 98.76% |
| 50% | 80% | 98.78% |
| 45% | 80% | 98.79% |
| 40% | 80% | 98.79% |
| 35% | 80% | 98.77% |
| 30% | 80% | 98.72% |
| 25% | 80% | 98.62% |
| 20% | 80% | 98.44% |
| 15% | 80% | 98.09% |
| 10% | 80% | 97.33% |
| 5% | 80% | 94.99% |

## CURVA DE PERFORMANCE (EFICIENCIA) DEL TRANSFORMADOR TAG: 2850-TX-015




DELCROSA S.A. V°B° INGENIERIA JORGE L. MENDEZ GIRON

| | Fecha | Nombre | Revisión: |
|---|---|---|---|
| Elaborado | 14/03/2023 | Roberto Neyra | 0 |
| Revisado | 14/03/2023 | Jhonattan Villanueva | |
| Aprobado | 14/03/2023 | Jorge Méndez | |